WORKED SOLUTION

STEP III 2002, Question 13

Probability and Statistics • Minimum of Exponentials

STEP III 2002, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q13

The key idea

"At least one of them is less than x" is the complement of "both are at least x", and the probability of the second event factorises by independence. Exponential tails are so simple, $\mathrm{P}(X_i \geq x) = \mathrm{e}^{-\lambda_i x}$, that the product is again an exponential tail, and that is the whole of the proof about the minimum. The buses are then a direct application: the wait for the next bus is the smaller of two exponential gaps.

Solution

The probability that one or both is less than x

First the tail of a single exponential variable. For $x \geq 0$,

$$\mathrm{P}(X_i \geq x) = \int_x^\infty \lambda_i \mathrm{e}^{-\lambda_i t}\,\mathrm{d}t = \left[-\mathrm{e}^{-\lambda_i t}\right]_x^\infty = \mathrm{e}^{-\lambda_i x}.$$

Let E be the event that $X_1 < x$ or $X_2 < x$ (or both). Its complement is the event that *both* $X_1 \geq x$ and $X_2 \geq x$. Since X_1 and X_2 are independent,

$$\mathrm{P}(X_1 \geq x \text{ and } X_2 \geq x) = \mathrm{P}(X_1 \geq x)\,\mathrm{P}(X_2 \geq x) = \mathrm{e}^{-\lambda_1 x}\mathrm{e}^{-\lambda_2 x},$$

so, for $x \geq 0$,

$$\boxed{\mathrm{P}(E) = 1 - \mathrm{e}^{-(\lambda_1+\lambda_2)x}.}$$

(For $x < 0$ the probability is 0, since neither variable can be negative.)

Ask yourself Why not just add $\mathrm{P}(X_1 < x)$ and $\mathrm{P}(X_2 < x)$?

Because the events overlap: both can be less than x at once, and adding would count that case twice. You can do it by inclusion–exclusion, $(1 - \mathrm{e}^{-\lambda_1 x}) + (1 - \mathrm{e}^{-\lambda_2 x}) - (1 - \mathrm{e}^{-\lambda_1 x})(1 - \mathrm{e}^{-\lambda_2 x})$, which simplifies to the same answer, but the complement is quicker.

The minimum is exponential

Let $X = \min(X_1, X_2)$. The smaller of two numbers is less than x exactly when at least one of them is less than x, so $\{X < x\}$ is the event E above. Hence the distribution function of X is

$$F(x) = \mathrm{P}(X \le x) = 1 - \mathrm{e}^{-(\lambda_1+\lambda_2)x} \qquad (x \ge 0),$$

with $F(x) = 0$ for $x < 0$. (For a continuous variable $\mathrm{P}(X < x) = \mathrm{P}(X \le x)$.) Differentiating, the density of X is

$$f(x) = F'(x) = (\lambda_1 + \lambda_2)\,\mathrm{e}^{-(\lambda_1+\lambda_2)x} \qquad (0 \le x < \infty),$$

which is the exponential density with parameter $\lambda_1 + \lambda_2$. So X has an exponential distribution, as required.

The buses

An exponential distribution with parameter λ has mean $\int_0^\infty t\lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1/\lambda$ (integrating by parts). Measuring time in minutes, the mean gaps of 15 and 30 minutes give parameters

$$\lambda_A = \tfrac{1}{15}, \qquad \lambda_B = \tfrac{1}{30}.$$

A bus on each route has just left, so the time until the next Route A bus is a full gap T_A between Route A buses, and likewise T_B for Route B. These are independent, exponential with parameters λ_A and λ_B. Either route takes me to college, so my wait is $W = \min(T_A, T_B)$, and by the result just proved W is exponential with parameter

$$\lambda_A + \lambda_B = \tfrac{1}{15} + \tfrac{1}{30} = \tfrac{1}{10}.$$

The median m satisfies $\mathrm{P}(W \le m) = \frac{1}{2}$:

$$1 - \mathrm{e}^{-m/10} = \tfrac{1}{2} \iff \mathrm{e}^{-m/10} = \tfrac{1}{2} \iff m = 10\ln 2.$$

Since $\ln 2 \approx 0.693$, the median wait is $m \approx 6.93$ minutes, which is approximately 7 minutes, as required.

Checks

- If $\lambda_2 \to 0$ (Route B buses almost never come), the answer becomes $1 - \mathrm{e}^{-\lambda_1 x}$, just the distribution of X_1, as it should.
- The mean wait is 10 minutes, shorter than either route's mean of 15 or 30, and the median $10\ln 2$ is below the mean, as usual for a right-skewed distribution.
- A simulation of 2×10^5 pairs of independent exponential gaps with means 15 and 30 gave a median minimum of about 6.91 minutes, against $10\ln 2 = 6.931\ldots$.

Common slips

- Using the means as the parameters: an exponential with mean 15 has $\lambda = \frac{1}{15}$, not 15.
- Adding the two probabilities $\mathrm{P}(X_1 < x) + \mathrm{P}(X_2 < x)$, which double-counts and can exceed 1.
- Averaging the routes' medians, or taking the median of route A alone ($15\ln 2 \approx 10.4$): having two routes shortens the wait.

Beyond STEP

Exponential gaps between buses mean the buses arrive as a *Poisson process*, and the result here says that merging two independent Poisson processes gives another one whose rate is the sum of the rates. Together with the memoryless property, $\mathrm{P}(X > s + t \mid X > s) = \mathrm{P}(X > t)$, this is the foundation of queueing theory and continuous-time Markov chains, met in first-year probability courses. Grimmett and Welsh, *Probability: An Introduction*, covers exponential variables and the Poisson process.

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